

TUAN PHAM

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SUMMARY

Computational Biologist & Machine Learning Researcher PhD Candidate with 4+ years of experience leveraging **Graph Neural Networks (GNNs)**, **Deep Learning**, and **multimodal integration** to decode gene regulation and neurogenomics. Proven expertise in developing scalable pipelines for **single-cell** (scRNA-seq, ATAC-seq) and **spatial transcriptomics**, with novel algorithms improving data alignment by 30%. Published author in *Nature Neuroscience* and *ACS Synthetic Biology* with a history of validating computational predictions in experimental settings.

EDUCATION

Brown University

PhD Candidate in Computational Biology

Thesis: Gene Regulation using Integrative Multimodal Modeling with Graph Neural Networks

Providence, RI

expected May 2027

GPA: 4.00

University of Rochester

BS in Computational Biology - Honors in Research

Minor in Computer Science

Rochester, NY

May 2022

GPA: 3.97

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, Bash
- **ML/AI:** PyTorch, TensorFlow, scikit-learn, GNNs, deep learning, Transformers, LLMs
- **Bioinformatics:** RNAstructure, Scanpy, Seurat, Bowtie2, SCENIC, SCOT, NGS
- **Tools & Systems:** Git, Linux, SLURM, Docker

SELECTED RESEARCH EXPERIENCE

Brown University

Graduate Researcher – Deep Learning & Single-Cell Genomics

Providence, RI

Sep 2022-now

- Developed **DRIFT**, a robust **graph diffusion-based** method for spatial transcriptomics **foundation models**, improving cell type annotation (**+20%**), spatial alignment (**+30%**), and clustering performance (**+20%**).
- Implemented **GRIDOT**, a **graph neural network** framework integrating **temporal causality** and **optimal transport** to model dynamic gene regulation; achieved **20% higher accuracy** than leading benchmarks on multi-omic single-cell datasets. Manuscript in prep.
- Applied **graph neural networks** and **optimal transport** to model gene regulation of odor-associated learning in temporal scRNA-seq datasets from *Mus musculus*. Findings predict candidate targets in PTSD-related circuits. Manuscript in prep.
- **Integrated scRNA-seq and ATAC-seq** data to prioritize synaptic TFs from roughly 1,000 candidates in *Drosophila* neurodevelopment - predictions were **experimentally validated**. (BioRxiv preprint)
- Inferred **gene trajectories** of 10,000 genes in the piriform cortex during mouse neuronal differentiation using OT-based alignment and developmental trajectory modeling (results published in **Nature Neuroscience**)
- **Mentored** four undergraduates in developing independent gene regulation analysis pipelines, leading to **two conference posters** and **three manuscripts**. Mentees were later accepted to **top graduate programs** including Harvard, Yale, and the Tri-Institutional Program.

University of Rochester

Undergraduate Researcher – Structural Biology & Genomics

Rochester, NY

Jan 2020-May 2022

- Designed 200 **DNA** and **RNA secondary structures** *in silico* with additional synthetic base pairs with RNAstructure with implication in synthesizing vaccines or aptamers, leading to a **first-author** publication in **ACS Synthetic Biology**.
- Developed a new method leveraging **NGS data** and sequence similarity networks to validate repetitive DNA homology across divergent genomes, presented results at the GLAM EVON Conference.

LEADERSHIP/TEACHING EXPERIENCE

Harvard University

Research Mentor

Boston, MA

Summer 2025

- Partnered with Harvard OpenBio to deliver COVID-focused research training for promising high school students; awarded **2nd Prize**, resulting in a **pending publication**.

Brown University

Providence, RI

Neurogenetics Teaching Assistant

Spring 2025

- Provided one-on-one support and strategic guidance during office hours, fostering analytical thinking and problem-solving skills.
- Facilitated collaborative learning environments to encourage peer-to-peer engagement and feedbacks for homework assignments and final projects.

International Student Mentor

2023-now

- Meno over **60** international graduate students in accessing academic and community resources, promoting a smooth transition to campus life.
- Fostered inclusive, supportive environments to enhance student engagement and curiosity.

Rochester Alpha Phi Omega (National Co-ed Service Fraternity)

Rochester, NY

Vice President of Service

2020-2022

- Coordinated and executed **1,200+ hours** of service for **60+ members** in partnership with **40+** Greater Rochester-based nonprofits, including Habitat for Humanity, Army Reserves, and Golisano Children's Hospital.
- Designed and executed fundraising initiatives (**over \$2,500**) for local nonprofits, including Make-A-Wish and House of Mercy, driving increased donor participation and expanded community outreach.

AWARDS AND HONORS

- T32 NIH BiOCN Fellowship 2025-2027
- Phi Beta Kappa (inducted 2021)
- Dean's Scholarship Recipient 2018-2022
- Student Employee of the Year Nominee 2021
- Rigby Wile Prize Recipient 2019 (annual award for a first-year class that shows proficiency in Biology)
- First Prize in Biomedical Data Science Hackathon Fall 2020 and 2021 (Undergraduate Division)

PUBLICATION

- Zeppilli, Sara, Alonso O Gurrola, Pinar Demetci, David H Brann, **Tuan M Pham**, Robin Attey, Noga Zilkha, et al. 2025. "Single-Cell Genomics of the Mouse Olfactory Cortex Reveals Contrasts with Neocortex and Ancestral Signatures of Cell Type Evolution." *Nature Neuroscience*, April <https://doi.org/10.1038/s41593-025-01924-3>.
- Colin Baker, **Tuan Pham**, Pinar Demetci, Quang Huy Tran, Ievgen Redko, Bjorn Sandstede, and Ritambhara Singh. 2025. "SCOT+: A Comprehensive Software Suite for Single-Cell alignment Using Optimal Transport." May. <https://doi.org/10.1101/2025.05.21.655322>
- **Pham, Tuan M.**, Terrel Miffin, Hongying Sun, Kenneth K. Sharp, Xiaoyu Wang, Mingyi Zhu, Shuichi Hoshika, et al. 2023. "DNA Structure Design Is Improved Using an Artificially Expanded Alphabet of Base Pairs Including Loop and Mismatch Thermodynamic Parameters." *ACS Synthetic Biology* 12 (9): 2750–63. <https://doi.org/10.1021/acssynbio.3c00358>.

PREPRINT

- Kentro, James A., Gunjan Singh, **Tuan M. Pham**, Justin Currie, Saniya Khullar, Audrey T. Medeiros, Maria Tsiarli, Erica Larschan, and Kate M. O'Connor-Giles. 2024. "Conserved Transcription Factors Coordinate Synaptic Gene Expression through Repression," October. <https://doi.org/10.1101/2024.10.30.621128>.
- Atishay Jain, **Tuan M. Pham**, David H. Laidlaw, Ying Ma, Ritambhara Singh. 2025 "Diffusion-based Representation Integration for Foundation Models Improves Spatial Transcriptomics Analysis." October. <https://doi.org/10.1101/2025.11.20.689624>

